

ModalPlayground: Collaborative Spatial Design of Multi-modal Haptic Feedback

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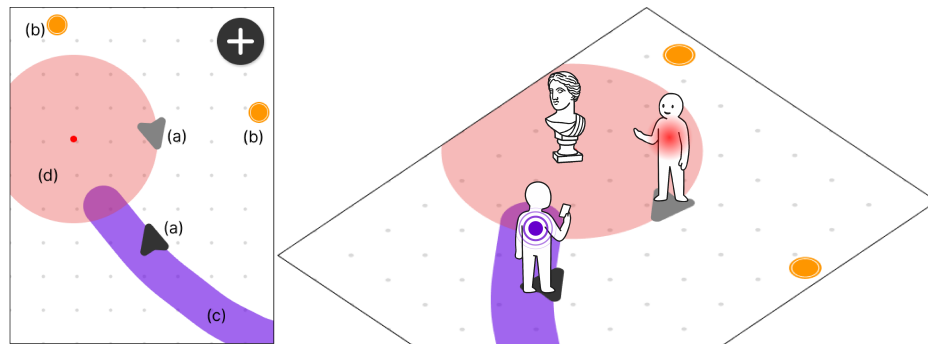


Figure 1: An overview of the ModalPlayground system. Users (a) localized via Ultra-wideband (UWB) anchors (b), can directly author and experience haptic zones on a physical canvas. They can "paint" different modalities, such as vibrotactile (c) or thermal (d) feedback, using their body as an input.

Abstract

We propose ModalPlayground, a collaborative design tool for authoring rich, multi-modal haptic feedback directly onto a physical space. Location-specific haptic feedback design faces significant barriers for creating immersive experiences that respond to users' positions in real-world spaces without tethering them to XR environments. The authoring process is often disconnected from the physical experience, leading to a gap between designer intent and user perception. ModalPlayground allows designers to use their own bodies as a 'paintbrush', physically walking through a space to define and layer haptic zones. Using high-precision Ultra-wideband (UWB) indoor positioning and the multi-modal haptic feedback capabilities of the DataFeel DevKit, this tool enables multiple designers to co-create and modify a shared haptic canvas in real time. This embodied, in-situ approach targets applications in immersive and interactive storytelling as well as spatial guidance.

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UIST Adjunct '25, Busan, Republic of Korea

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ACM ISBN 979-8-4007-2036-9/25/09

<https://doi.org/10.1145/3746058.3758977>

CCS Concepts

• **Human-centered computing** → **Haptic devices**.

Keywords

Multi-modal haptics, Location-based feedback, Collaborative design

ACM Reference Format:

Louis Badr and Thomas Juldo. 2025. ModalPlayground: Collaborative Spatial Design of Multi-modal Haptic Feedback. In *The 38th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '25)*, September 28–October 01, 2025, Busan, Republic of Korea. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3746058.3758977>

1 Introduction and Related Work

Imagine walking through a museum exhibit about ancient Rome and feeling a sense of warmth as you approach a display on Roman baths, or following a treasure map where the path forward is a gentle, pulsing sensation on your wrist. Although the potential for such experiences is vast, creating location-specific haptic feedback is currently challenging. Current methods mainly rely on XR software development kits, while some approaches are oriented toward beacons [2] or GPS-based mapping and localization services for spatial guidance [1]. Emerging collaborative design approaches for haptic feedback are beginning to address these limitations [3], yet

they often remain tethered to digital interfaces rather than physical spaces. This is where our work diverges: we move the design process entirely out of the screen and into the physical world. New wearable technologies for sub-meter precision tracking now exist and allow freedom from XR constraints but designing feedback would still require programming coordinates on maps, a process that is abstract and disconnected from the final, embodied user experience. This creates a significant designer-user gap, where the designed effect feels fundamentally different in the complex, three-dimensional physical world. This project aims to bridge this exact gap. Our intuition is that to design for the body in a space, one must use the body in that space.

Our contributions are:

- An embodied, in-situ authoring paradigm: We introduce a "haptic painting" metaphor where designers physically walk through a space to define zones, using their tracked body as the input device.
- A real-time, collaborative haptic canvas: Our system allows multiple designers to work simultaneously in the same physical space, layering and refining haptic feedback together to create better experiences.
- A complete system implementation: We demonstrate a functional prototype integrating UWB indoor positioning with the multi-modal feedback capabilities (vibration, heat, cold, and light) of the DataFeel DevKit.

2 Design Space & Rationale

Our design is guided by two core principles aimed at closing the designer-user gap.

Embodied Design. Traditional design involves mapping coordinates. By treating the designer's body as a 'brush' and the physical space as a 'canvas,' we enable a more fluid, direct and embodied form of creation. Instead of locating a point on a map, a designer can go where he imagines the haptic feedback to start and stop. A designer can intuitively gauge the size of a zone by walking its perimeter or feel the intensity of a gradient by walking through it, ensuring the final effect matches their expectations.

Collaborative Authoring. Haptic perception is subjective, a sensation one designer finds intuitive may be confusing to another. ModalPlayground aims to address this by allowing multiple designers to simultaneously experience and modify the same haptic canvas. This transforms the design process into a conversation with rapid iterations, as the team can immediately feel the effects of each other's changes, discuss them, and build upon them on the spot.

3 Design Implementation

3.1 Hardware Architecture

The environment is equipped with DW1000 UWB modules that act as fixed anchors for real-time positioning. Each designer has a wearable node that consists of an El Jefe controller, a DataFeel Dot, and a UWB tag that communicates wirelessly with their personal smartphone. The smartphone runs the ModalPlayground application and serves as the primary control interface. The complete system architecture is shown in Figure 2.

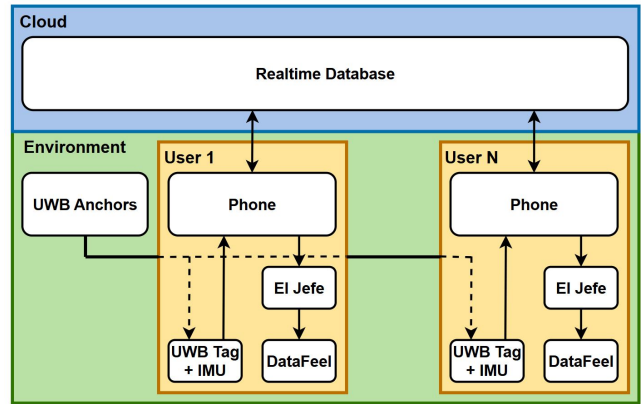


Figure 2: System architecture diagram of ModalPlayground

3.2 Software Architecture and Data Flow

Each designer's smartphone runs the ModalPlayground app. The app maintains two simultaneous Bluetooth connections: one to the El Jefe™ controller to send haptic commands, and one to the UWB tag to receive real-time position data (x,y). The app processes the incoming position data. In "Design Mode," the user uses the phone's touch interface to select a haptic "brush" (e.g., "Warmth Gradient"). As they walk, the app records their position and translates it into a haptic zone. All haptic zone data (creations, modifications, deletions) is instantly synchronized with a cloud-based real-time database. The app on each designer's phone subscribes to updates from the cloud database. When one designer creates a new zone, the data is sent to the cloud and is immediately pushed to all other designers' phones. This ensures every participant shares the exact same, constantly updated canvas.

4 Application Scenarios

We envision multiple use cases for ModalPlayground centered on spatial guidance and immersive storytelling. In cultural contexts, ModalPlayground could enhance museum visits and art exhibitions, enabling curators to rapidly prototype and test haptic experiences even when physical layouts change. For entertainment, both home-based and professional scenarios emerge, such as treasure hunts, escape games, or scenarios where the design of haptic maps becomes part of the gameplay itself. ModalPlayground could also support accessibility applications, potentially providing navigation assistance for visually impaired users. It could serve as an awareness tool, allowing designers to explore empathy-building experiences that simulate different sensory conditions for educational purposes.

Acknowledgments

Funded/Co-funded by the European Union (ERC, KinestheticDisplays, 101165100). Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Research Council. Neither the European Union nor the granting authority can be held responsible for them.

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